Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Provisioned and operated multi-cloud **GPU clusters** (AWS, GCP, CoreWeave) with **NVIDIA/CUDA** and **InfiniBand**, running distributed training on **Slurm** and **Kubernetes**.
- Built the **evaluation harnesses** that gate model and agent releases, scoring quality, **hallucination**, and grounding with **regression suites**, plus **vLLM** serving observability.
- Ran **distributed training** and fine-tuning of **PyTorch** models on multi-GPU clusters, scheduling jobs with **Slurm** and **Kubernetes**.
- Led a production **multi-agent system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** tool use that autonomously processes **financial workflows**, handling **millions of daily transactions** at **99%+ uptime**.

QureAI | *ML Infrastructure Engineer* Mar 2026 – Present

- Moved LLM serving to an **async, autoscaled** architecture with **TensorRT-LLM**, cutting **p99 latency from 420ms to 140ms** across **50K+ daily requests**.
- Worked on **vLLM-class** serving internals including distributed **KV cache** and **speculative decoding** to raise inference throughput.
- Built **PySpark** jobs and **Airflow** DAGs for large-scale feature and dataset preparation feeding training and **evals**.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime** on **AWS**.
- Cut nightly batch aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** on a financial RDBMS with indexed views and partitioning.
- Improved computer-vision model accuracy **9.6 points** and cut false negatives **31%** by evaluating **CNN, ResNet, UNet, and ViT** architectures and tuning decision thresholds.
- Cut model regressions **28%** and validation time from **6 hours to 90 minutes** with an automated **model-evaluation and deployment-gating** framework.

Projects

Road Lane Detection (M.S. Thesis, UT Arlington) `PyTorch`

- Built and benchmarked **lane segmentation** models for road scenes, comparing **UNet**, **CNN**, **Vision Transformer (ViT)**, and **Swin Transformer** architectures in **PyTorch**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- **LLM evaluation** platform with rubric-based scoring, **hallucination detection**, regression alerting, a **FastAPI** backend, and a **React/TypeScript** dashboard.

HoG-ViT `github.com/ganeshhgupta/HoG-ViT`

- **Vision Transformer** that folds **Histogram of Oriented Gradients (HOG)** features into patch embeddings for **image classification**.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Aug 2016 – Jul 2020

Skills

Languages: Python, SQL, Java, TypeScript

ML Frameworks: PyTorch, JAX, TensorFlow, Hugging Face, vLLM, LoRA/QLoRA, Quantization

Research to Production: Model Training and Fine-Tuning, Evaluation Frameworks, Multi-GPU Training, Model Serving, Deployment Gating

Infrastructure: AWS (SageMaker, Lambda, S3), GCP (Vertex AI), Docker, Kubernetes, Langfuse, CI/CD

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer